Models as Parametric Kernels, Governance as Verified Automation:

A Mathematical Foundation for Heterogeneous Model Estates and a Criterion for Where AI May Do the Work

Ashutosh Sinha

Independent Researcher

ajsinha@gmail.com

Abstract

Large regulated organisations operate thousands of quantitative artefacts that inform or make consequential decisions: closed-form valuation formulae, market-calibrated stochastic models, statistically estimated scorecards, machine-learned classifiers, online-adaptive systems, foundation-model applications, third-party black boxes, expert-judgment rating schemes, and deterministic rule sets. Supervisory regimes require that these be identified, classified by materiality, validated, monitored, and documented as a single population. Yet the software systems built to discharge that obligation divide sharply into governance tools that record assertions *about* artefacts without access to them, and engineering platforms that manage artefacts without any representation of materiality, approved use, independent challenge, or remediation. We argue that this division is not an accident of product strategy but a symptom of a missing foundation: the field has no formal answer to the question *what is a model?* that is simultaneously general enough to cover the population and structured enough to support composition, substitution, extension, and plural normative interpretation.

We propose such a foundation. A model is a morphism in **Para**(**Stoch**): a parameter object together with a Markov kernel consuming parameters and input to produce output. We show that the familiar informal taxonomy of model kinds is recovered not as a partition of objects but as a classification of *how the parameter object is inhabited*, which dissolves the recurring difficulty that many governed artefacts are never trained. We prove that no aggregate risk assignment can be compositional, and identify the obstruction precisely as the comonoid copy map — shared dependency — thereby giving formal content to a longstanding but informally stated supervisory expectation. We model artefact kinds as fibres of a fibration and normative regimes as institutions in the sense of Goguen and Burstall, obtaining conservative-extension guarantees for both, and we show that the institutional satisfaction condition yields a falsifiable consistency test for regulatory scope determinations. Finally we show that assurance evidence forms an annotated derivation structure over a commutative semiring, so that questions of sufficiency, minimal justification, confidence, cost, classification, admissibility, and currency are one computation evaluated in different semirings.

The contribution is conceptual and integrative rather than deep: each mathematical instrument is borrowed, and we are explicit about which. What is new is their assembly into a coherent theory of a domain that presently has none, together with five results that this assembly makes available. We state the limitations candidly, including the absence of an implementation study, and identify what would falsify the account.

Keywords: model governance; categorical probability; Markov categories; institutions; provenance semirings; assume–guarantee contracts; fibrations; model risk.

1 Introduction

1.1 The domain

Consider the population of quantitative artefacts operated by a large financial institution. It includes a Black–Scholes implementation whose parameters come from financial theory and are never estimated from data; a stochastic volatility surface re-solved against market quotes every morning; a logistic scorecard refitted annually on a historical sample; a gradient-boosted classifier retrained weekly on transaction streams; an adaptive threshold that updates itself continuously in production; a retrieval-augmented language model drafting regulatory narratives; a third-party credit score whose internals are contractually unavailable; a country-risk rating scheme whose weights are set by a committee; and several thousand rule sets and spreadsheets.

These artefacts have essentially nothing in common at the level of implementation. Yet they are subject to a single normative obligation. Supervisory guidance requires an institution to maintain an inventory of them, classify each by materiality, subject each to independent challenge proportionate to that classification, monitor performance, document design and limitations, control change, and assess risk both individually and *in aggregate*, accounting for "interactions and dependencies among models; reliance on common assumptions, data, or methodologies." Different jurisdictions define the governed population differently: one recent regime narrows it to complex quantitative methods applying statistical, economic or financial theory, explicitly excluding deterministic rules and, remarkably, generative systems; another defines it broadly enough to include expert-judgment inputs and qualitative outputs. An institution operating in both must satisfy a narrow and a broad scope simultaneously over one population.

1.2 Four pathologies

Systems built for this task exhibit four recurring pathologies, each of which we will argue is a consequence of an absent formalism rather than of poor engineering.

P1: The training assumption. Tooling descended from machine learning practice presumes a pipeline train → register → deploy. Artefacts that are not trained — which are the majority — are represented as degenerate cases, with mandatory fields left empty and evidence requirements that are category errors. Asking a closed-form pricer for its training dataset is not a missing value; it is a type error, and a system that cannot say so will accumulate meaningless records.

P2: Schema rigidity. When artefact kind is a column, an enumeration, or a table, each new kind requires a schema migration touching every module. The predictable consequence is that new kinds are never added and the inventory silently excludes them — precisely the artefacts most likely to be novel and therefore least understood.

P3: Scope as a boolean. When regulatory applicability is modelled as an attribute rather than as a distinct logic, overlapping regimes cannot be represented without conflict, and every regulatory revision forces re-architecture. This is not hypothetical: a fifteen-year-old supervisory letter that anchored an entire industry's practice was superseded in 2026 with a materially different definition of the governed population.

P4: Evidence as assertion. Governance records state that an artefact was validated. Nothing in the record establishes that the artefact described is the artefact executing. The claim is unfalsifiable, which is the opposite of what assurance requires.

1.3 Contributions

We give a mathematical foundation addressing all four. Concretely:

1. Section 2 defines a model as a morphism in **Para**(**Stoch**) and shows that the standard informal taxonomy is recovered as a classification of *fitting morphisms* $\varphi\colon D \to P$ into the parameter object (Proposition 2.3). This dissolves P1: training is one way of inhabiting a parameter object, not part of the definition of a model.
2. Section 3 proves that no aggregate risk assignment can be compositional (Theorem 3.3), and identifies the obstruction as the comonoid copy map. This gives formal content to the supervisory expectation regarding interactions and common dependencies, and explains why Markov categories — which make copying explicit — are the right ambient setting rather than Cartesian ones, which hide it.
3. Section 4 treats artefacts that cannot be opened, combining assume–guarantee contracts, sound abstraction in the sense of abstract interpretation, and a *probe-relative* weakening of the Yoneda principle. We obtain a soundness criterion for summary documents (Proposition 4.1) and a formal semantics for version increments (Proposition 4.3).
4. Section 5 models artefact kinds as fibres of a fibration and proves a conservative extension property (Proposition 5.1), addressing P2.
5. Section 6 models normative regimes as institutions and shows that the satisfaction condition yields a falsifiable consistency test for scope determinations (Proposition 6.1), addressing P3.
6. Section 7 presents assurance evidence as an annotated derivation structure over a commutative semiring, so that seven distinct governance questions are one computation evaluated in different semirings (Proposition 7.1), addressing P4.
7. Sections 8 and 9 treat risk classification as a monotone map between lattices adjoint to control intensity, and give a characterisation of causally admissible training assemblies over a bitemporal store (Proposition 9.2).
8. Section 11 draws a consequence we did not anticipate and now regard as the most practically significant: several of the constructions above are *decision procedures*, and a decision procedure is what makes machine-generated output safe to accept (Proposition 11.3). We show that verifying a machine-generated citation reduces to a Boolean evaluation over the existing derivation structure (Proposition 11.4), identify which governance tasks are oracle-backed and which constitutively cannot be (Observation 11.5), and show that a governing system may use governed models without circularity provided generation crosses the strata and acceptance does not.

Section 10 observes that every result above can be stated as an executable law and enforced by property-based testing, which we regard as the practical criterion separating a foundation from an ornament. Section 12 works a small but complete example. Sections 13 and 14 situate the account and state candidly what it does not establish.

1.4 How to read this paper

The mathematics is elementary but the vocabulary is not evenly distributed across the audiences this work is addressed to. Every formal statement is therefore followed by two things: a shaded *plain terms* gloss that says what the statement means in ordinary language, and, where it earns one, a *worked example* drawn from banking. A reader who wants the argument without the notation can follow the shaded material alone and lose nothing essential. A reader who wants only the mathematics can skip it.

1.5 What is borrowed and what is new

We are explicit, because the value of an integrative paper depends on it. Markov categories are due to Fritz and collaborators; the **Para** construction to Fong, Spivak and Tuyeras, and its elaboration to Cruttwell, Gavranović, Lehner, Wilson and co-workers; assume–guarantee contract algebra to Benveniste and collaborators; institutions to Goguen and Burstall; provenance semirings to Green, Karvounarakis and Tannen; sound abstraction to Cousot and Cousot; well-behaved lenses to Foster and collaborators; sheaf-theoretic consistency measures to Robinson. None of these is our contribution.

What we claim is new: (i) the identification of **Para**(**Stoch**) as the right ambient setting for a *governed population* of heterogeneous artefacts, and the resulting reading of the trainability taxonomy; (ii) Theorem 3.3 and its diagnosis of the copy map as the obstruction; (iii) the application of institution theory to plural regulatory scope over a shared inventory, with the satisfaction condition repurposed as a consistency test; (iv) the application of semiring provenance to assurance evidence rather than to query answers, and the resulting collapse of seven governance questions into one; (v) the probe-relative treatment of artefact identity and its use as a semantics for version increments; and (vi) the oracle analysis of Section 11, including Proposition 11.4 and the constitutivity argument of Observation 11.5.

2 Models as parametric kernels

2.1 Ambient setting

We work in a Markov category: a symmetric monoidal category $(\mathcal{C}, \otimes, I)$ in which the unit I is terminal and every object X carries a commutative comonoid structure $\mathrm{copy}_X : X \to X \otimes X$, $\mathrm{del}_X : X \to I$, compatible with the monoidal product, with del natural. Morphisms behave as Markov kernels; the canonical example is **Stoch**, the Kleisli category of the Giry monad on measurable spaces, whose morphisms $X \to Y$ are probability kernels. A morphism f is *deterministic* when $\mathrm{copy}_Y \circ f = (f \otimes f) \circ \mathrm{copy}_X$.

Two structural features earn their place immediately. Terminality of I (semicartesianness) says every morphism may be discarded — outputs need not be used, which is what makes an artefact's *exposure* a meaningful separate quantity. Explicit copying says an output may be consumed by several downstream artefacts — which, as Section 3 shows, is exactly the structure responsible for non-compositional aggregate risk. A Cartesian category has both properties implicitly and therefore cannot make either visible.

2.2 Definition

Definition 2.1 (Model)**.** Let $\mathcal{C}$ be a Markov category. A *model* is a morphism in **Para**($\mathcal{C}$): a pair (P, f) consisting of a *parameter object* $P \in \mathcal{C}$ and a kernel

$$f : P \otimes X \longrightarrow Y,$$

where X is the *input object* and Y the *output object*. Composition of $(P, f) : X \to Y$ and $(Q, g) : Y \to Z$ is $(Q \otimes P,\ g \circ (\mathrm{id}_Q \otimes f))$.

> **In plain terms.** A model is a machine with *dials*. X is what you feed in, Y is what comes out, and P is the set of dials that determine how the machine behaves. The definition insists on keeping the dials separate from the machinery, because that separation is where all the leverage is.
> The word *kernel* means the output is allowed to be random: feed in the same input twice and you may get different answers. That covers a language model sampling text. Ordinary deterministic behaviour is the special case where the randomness happens to be trivial.
> The composition rule says: if you plug one machine into another, the combined machine's dials

are *both* sets of dials together. Nothing is lost or absorbed.

Worked example: two artefacts that look nothing alike

A **Black–Scholes option pricer** takes a spot price, strike, tenor, rate and volatility, and returns a price. Its formula comes from theory. There are no dials to set — P is empty. Feed it the same inputs and you get the same price forever.

A **SABR volatility surface** takes a strike and tenor and returns an implied volatility. But before it can do that, four parameters must be solved for against this morning's market quotes. Those four numbers are P, and they are different every day.

Same shape, completely different operational reality — and the difference lives entirely in P.

Definition 2.2 (Fitting morphism)**.** A *fitting morphism* for a model (P, f) is a morphism $\varphi\colon D \to P$ in $\mathcal{C}$, where D is an object of *fitting evidence*. A model is *accompanied* when a fitting morphism is available to the governing party, and *opaque* otherwise.

In plain terms. The *fitting morphism* is the procedure that sets the dials. D is whatever the procedure consumes: a historical sample, this morning's market quotes, a committee's opinions, a prompt and a document corpus.

The key move — and the one that departs from received practice — is that *setting the dials is not part of being a model.* It is a separate arrow. Some models have one. Some do not. Some have one that belongs to a vendor and that you will never see.

"Opaque" is the precise term for a bought-in model: the dials exist, somebody set them, and you cannot inspect either.

Definition 2.2 is where the account departs from received practice. Estimation, calibration, training, elicitation and configuration are not different *kinds of model*; they are different fitting morphisms into the parameter object of one kind of thing.

2.3 The trainability classification

Proposition 2.3 (Classification)**.** *The informal taxonomy of governed artefact kinds is recovered as a classification of the pair (P, φ), as follows.*

Class	***Parameter object P***	***Fitting morphism φ***	***Determinism of f***
Analytic	$P \cong I$ *(terminal)*	*none required*	*deterministic*
Calibrated	*calibration parameters*	*solver on an instrument set*	*deterministic given P*
Estimated	*coefficients*	*estimator on a sample*	*often deterministic*
Learned	*weights, hyperparameters*	*training algorithm*	*varies*
Adaptive	*time-indexed $P(t)$*	*φ is itself a process*	*non-stationary*
Foundation-based	*weights, prompt, corpus, tools*	*configuration and retrieval*	*stochastic*
Opaque	*inaccessible*	*external, unavailable*	*unknown*
Elicited	*weights or thresholds*	*elicitation protocol*	*deterministic given P*
Authored	*rule set*	*policy authoring*	*deterministic*

Proof. Each row is a specification of (P, φ) within Definitions 2.1 and 2.2; that the rows are exhaustive of current practice is an empirical claim about the domain, not a mathematical one, and we make it as such. The two extremal cases are the mathematically interesting ones. *Analytic* is the case $P \cong I$: since I is terminal, $I \otimes X \cong X$ and the model is a bare kernel $X \to Y$, with the fitting problem vacuous. *Opaque* is the case in which P exists but neither P nor φ is available to the observer, so that the only accessible datum is the composite $f(p, -)$ for the vendor's fixed p; that composite is an ordinary morphism $X \to Y$, which is precisely why behavioural evidence is the only evidence obtainable. $\square$

In plain terms. Practitioners argue endlessly about whether a pricer "is a model", whether a rule set counts, whether a chatbot belongs in the inventory. The table says the argument is misconceived. These are not nine different kinds of thing. They are one kind of thing, distinguished by *how the dials get set.*
The two ends of the table are worth dwelling on. At the top, the dials do not exist, so asking about them is meaningless. At the bottom of the opaque row, the dials exist but are behind a contract, so asking about them is futile. Everything else is in between.

Worked example: three artefacts through the same lens

An SA-CCR counterparty exposure calculator. The formula is prescribed by regulation. $P \cong I$. There is no dataset, no estimator, no retraining schedule. What you must evidence is that the implementation computes what the regulation says — a test suite against reference values, not a validation sample. Any system demanding a "training set" here is asking a meaningless question, and will get a meaningless answer typed into a mandatory field.
A small-business PD scorecard. P is forty-two coefficients. φ is a logistic regression run on eight years of originations. Evidence: the sample, its representativeness, the estimator, out-of-time performance, calibration.
A bought-in bureau score. P exists — somebody at the bureau fitted it — but neither P nor φ is available. Demanding development documentation is futile. What you can do, and what the mathematics says is the *only* thing you can do, is watch the composite: compare its outputs against your own realised defaults. That is not a compromise forced by commercial reality. It is what the structure permits.

Remark 2.4. Proposition 2.3 is the resolution of pathology P1. There is no "untrained model" and no "degenerate case"; there is a model whose parameter object is terminal. The design consequence is that evidence requirements are indexed by the classification rather than assumed uniform — formalised in Section 5.

Proposition 2.5 (Parameter accumulation)**.** *For composable models $(P, f)\colon X \to Y$ and $(Q, g)\colon Y \to Z$, the composite has parameter object $Q \otimes P$. Consequently, in any finite network the composite parameter object is the tensor of the parameter objects of all constituents reachable from the inputs.*

Proof. Immediate from composition in $\mathbf{Para}(\mathcal{C})$ and induction on network size. □

Corollary 2.6 (Change closure)**.** *Any change to the inhabitant of a constituent parameter object changes the inhabitant of the composite parameter object. Hence recalibration of an upstream artefact is, formally, a change event for every downstream artefact.*

This corollary is worth pausing on: it is routinely violated in practice, where a curve is recalibrated daily and the dozens of valuation and capital artefacts consuming it are held to be unchanged.

In plain terms. If you build something out of other things, you inherit *all* of their dials. A valuation chain four models deep has the dials of all four. So when somebody turns a dial anywhere upstream, the thing at the end of the chain has genuinely changed — even though nobody edited it.

Worked example: the daily recalibration nobody calls a change

At 6:30 each morning a bank's discount curve is re-solved against market instruments. Its parameter set changes. Downstream sit a swaption pricer, an XVA engine, a VaR engine, a capital calculator and a capital plan.
By Proposition 2.5, every one of those artefacts has had its parameter object change. Formally, each has changed. Operationally, the bank's change-management process records nothing: curves

are “market data”, not a model change.
Both readings are defensible, but they must be reconciled explicitly rather than by silence. The usual resolution is to distinguish routine recalibration *within a declared tolerance* — which is governed by monitoring — from recalibration outside it, which is a change event. The formalism does not choose for you; it forces you to notice that a choice is being made, and to write down the tolerance.

2.4 Output type

Models producing point estimates and models producing predictive distributions differ in the object Y, not in kind. In a *representable* Markov category, kernels $X \to Y$ correspond to deterministic morphisms $X \to \mathcal{P}Y$ into a distribution object; a point predictor is then the deterministic section obtained by post-composing with a summary statistic. This is what permits calibration assessment, interval-based monitoring and proper scoring rules to be defined once, generically, rather than per artefact family.

3 Composition and the impossibility of compositional risk

3.1 Networks as string diagrams

$\mathbf{Para}(\mathcal{C})$ is symmetric monoidal, so a network of models is a string diagram built from generators by sequential composition, parallel composition, copying and discarding. Type-correctness of a wiring is decidable, and the transitive downstream closure of a node — what practitioners call the blast radius of a change — is a graph-theoretic property of the diagram rather than a report maintained beside it.

3.2 Aggregate risk

Supervisory guidance asks that risk be assessed “both individually and in aggregate,” where aggregate risk “reflects interactions and dependencies among models; reliance on common assumptions, data, or methodologies.” We now show this expectation has a precise formal content: it asserts the failure of compositionality.

Definition 3.1 (Risk assignment)**.** Let R be a join-semilattice. A *risk assignment* is a function ρ from morphisms of $\mathbf{Para}(\mathcal{C})$ to R. It is *compositional* if for all composable f, g and all f, h,

$$\rho(g \circ f) = \rho(g) \sqcup \rho(f), \qquad \rho(f \otimes h) = \rho(f) \sqcup \rho(h).$$

Compositionality is the assumption implicitly made whenever an aggregate is computed as the maximum or join of constituent ratings — which is standard practice.

Definition 3.2 (Failure semantics and fragility)**.** Fix a network N with generator set $G(N)$ and output ports $\mathrm{out}(N)$. For $S \subseteq G(N)$ let $\mathrm{corr}(N, S) \subseteq \mathrm{out}(N)$ be the set of output ports reachable in the diagram from some generator in S. The *fragility* of N is $\mathrm{frag}(N) = \max_{g \in G(N)} |\mathrm{corr}(N, \{g\})|$. A risk assignment ρ is *fragility-sensitive* if for networks N, N' with $\mathrm{frag}(N) > \mathrm{frag}(N')$ we have $\rho(N) \sqsupset \rho(N')$ whenever the multisets $\{\{\rho(g) : g \in G(N)\}\}$ and $\{\{\rho(g') : g' \in G(N')\}\}$ coincide.

Theorem 3.3 (No compositional risk assignment)**.** *Let R be a join-semilattice with at least two elements. No risk assignment is both compositional and fragility-sensitive.*

Proof. Let $a \sqsubset b$ in R. Take generators $A, A' \colon X \to Y$ with $\rho(A) = \rho(A') = a$, and $B, C \colon Y \to Z$ with $\rho(B) = \rho(C) = b$; A and A' are distinct generators of equal risk (*e.g.* two independently constructed curve models).

Let N_1 be the network in which the output of A is copied and fed to both B and C; that is, $N_1 = (B \otimes C) \circ \mathrm{copy}_Y \circ A$, with two output ports. Let N_2 be the network $(B \circ A) \otimes (C \circ A')$, again with two output ports.

The multisets of constituent risks coincide: both are $\{\{a, b, b\}\}$. But $\mathrm{corr}(N_1, \{A\}) = \mathrm{out}(N_1)$, so $\mathrm{frag}(N_1) = 2$, whereas every generator of N_2 reaches exactly one output port, so $\mathrm{frag}(N_2) = 1$.

Now suppose ρ is compositional. Then $\rho(N_1) = a \sqcup b \sqcup b = a \sqcup b$ and likewise $\rho(N_2) = a \sqcup b$, so $\rho(N_1) = \rho(N_2)$. This contradicts fragility-sensitivity, which requires $\rho(N_1) \sqsupset \rho(N_2)$. $\square$

> **In plain terms.** Two banks each run one curve model and two pricers. Bank A uses a single curve to feed both pricers. Bank B built two curves independently, one for each pricer.
> Rate their models individually and you get identical answers: same curve rating, same two pricer ratings. So any method that computes the total from the individual ratings gives both banks the same answer.
> But Bank A has a single point of failure and Bank B does not. If the curve is wrong at Bank A, both valuations are wrong. At Bank B, one is. The banks are plainly not equally exposed — and the standard method cannot see the difference, because the difference is not in any of the parts.
> That is the whole theorem: *you cannot get the risk of the whole by combining the risks of the parts, because the risk of the whole partly lives in the wiring.*

> **Worked example: why this is not a toy**
>
> At a real institution the shared node is not hypothetical. A single USD discount curve typically feeds swaption and swap pricers, the CDS and tranche pricers, the XVA engine, the VaR and expected shortfall engines, the FRTB standardised sensitivities, prudent valuation, and — through the RWA engine — the capital plan and the regulatory submission.
> Rate each of those individually and you might reasonably conclude the portfolio is well controlled: each is validated, each is monitored, each is within tolerance. Every constituent is green.
> Yet an error in curve construction propagates to every one of them simultaneously, in the same direction, and the errors do not offset — they compound, because the downstream artefacts all consume the same wrong number. This is exactly the concentration that the supervisory language about "reliance on common assumptions, data, or methodologies" is pointing at, and exactly what a maximum-of-components rating is structurally incapable of representing.
> The practical output is a number: report the join of the component risks, report the composite risk, and attribute the difference to the specific shared curve. That difference is the interaction premium of Corollary 3.5, and it is what tells a committee that the estate's real exposure is concentrated in one artefact that nobody has flagged, because individually it looks fine.

Remark 3.4 (The obstruction is the copy map). The two networks differ in exactly one respect: N_1 uses copy_Y and N_2 does not. Shared dependency *is* the comonoid copy map. In a Cartesian category copying is implicit and structurally invisible, so the distinction between N_1 and N_2 cannot be expressed at the level of the algebra; this is a concrete reason to prefer a Markov category as the ambient setting even when the artefacts under management are deterministic.

Corollary 3.5 (Laxity is necessary)**.** *Any fragility-sensitive risk assignment is at best* lax *monoidal: $\rho(g \circ f) \sqsupseteq \rho(g) \sqcup \rho(f)$, with the discrepancy attributable to shared constituents. We call $\rho(N) \ominus \bigsqcup_{g \in G(N)} \rho(g)$ the* interaction premium.

The practical reading of Corollary 3.5 is that a governance system should report the join of constituent risks *and* the composite risk, and should attribute the difference. Attribution proceeds by the sources of shared dependency admitted by the diagram: common parameter objects (Proposition 2.5), common inputs, common fitting evidence D, and common methodology. To our knowledge no deployed system computes this quantity, though every relevant supervisory regime asks for it in prose.

4 Specifying artefacts one cannot open

A substantial fraction of a governed population is opaque in the sense of Definition 2.2. Three complementary instruments apply, answering different questions.

4.1 Contracts

A *contract* is a pair $\mathcal{K} = (A, G)$ of an assumption on the environment and a guarantee on behaviour conditional on the assumption. Contract algebra supplies refinement $\mathcal{K}' \preceq \mathcal{K}$ (weaker assumption, stronger guarantee), composition, conjunction, and a quotient $\mathcal{K}/\mathcal{K}_1$ giving the obligation of a missing component.

The mapping into the domain is exact, and clarifies a term that supervisory texts use without definition. The *operating boundary* of a model — the region over which its performance is expected to be acceptable — is the assumption A: input domain, population definition, regime, freshness of upstream inputs. The performance envelope — discrimination, calibration, fairness, latency bounds — is the guarantee G. Two consequences follow. First, use outside the operating boundary is a contract violation, checkable at execution time; when A fails, G is void, which is a considerably stronger statement than "the input looked unusual." Second, substitutability becomes a proof obligation: replacing an artefact by another is admissible exactly when the replacement's contract refines the incumbent's. Under the standard variance discipline this requires contravariance in inputs and covariance in outputs.

> **In plain terms.** A contract is a promise with a precondition: *if* you use me within these limits, *then* I will perform this well. Outside the limits I promise nothing — and that is a much stronger and more useful statement than "results may be unreliable".
> The refinement rule answers "can version B replace version A?" precisely: B must accept everything A accepted (or more) and promise everything A promised (or more). That is checkable, so replacing a production model becomes a proof obligation rather than a judgement call in a meeting.
> The quotient operation is the one people find surprising: given what you need overall and what you already have, it computes the specification of the missing piece. A validation gap becomes a requirements document, automatically.

> **Worked example: an operating boundary that mattered**
> A small-business PD scorecard is fitted on 2014–2019 data. Its assumption A records, among other things, "non-recessionary conditions; unemployment below 8%". Its guarantee G records a discrimination floor and a calibration test.
> In April 2020, unemployment moves outside that range. Under the usual arrangement, nothing happens: the model keeps scoring, performance degrades over the following quarters, and the degradation is noticed at the next scheduled validation — perhaps nine months later, after several thousand decisions.
> Under a contract, the assumption is machine-checked on every call. The moment inputs fall outside A, the guarantee G is formally void. The system can flag, refer to a human, or refuse, according to policy set in advance — and the count of boundary violations is a live number on the model's page rather than a discovery made retrospectively.
> Note what the formalism contributes here. It does not tell you the model is wrong. It tells you the model has stopped promising anything, which is a different and more actionable fact.

4.2 Summaries as sound abstractions

Documents that stand in for artefacts — summary cards, risk statements, disclosed limitations — are abstractions, and abstract interpretation says what makes an abstraction trustworthy. Let $(\mathcal{B}, \sqsubseteq)$ be a lattice of behaviours and $(\mathcal{S}, \sqsubseteq)$ a lattice of summaries, with $\alpha\colon \mathcal{B} \to \mathcal{S}$ and $\gamma\colon \mathcal{S} \to \mathcal{B}$ forming a Galois connection $\alpha \dashv \gamma$.

Proposition 4.1 (Summary soundness). *A summary s of behaviour b is* sound *iff $b \sqsubseteq \gamma(s)$, equivalently $\alpha(b) \sqsubseteq s$. A sound summary may overstate the set of admissible behaviours — and hence overstate risk — but never excludes behaviour the artefact exhibits.*

Proof. Immediate from the Galois connection. □

Proposition 4.1 gives a testable criterion where practice offers only editorial judgement: a summary stating a discrimination floor is unsound if the artefact falls below it on any slice within its operating boundary, and the violation is detectable by replay. It also answers, principledly, the perennial question of how much detail a summary should contain: enough to remain sound at the required precision, and no more.

> **In plain terms.** A summary of a model is allowed to be pessimistic. It is not allowed to be flattering.
> If the model card says "discrimination at least 0.82" and the model achieves 0.79 anywhere inside its declared operating boundary, the card has understated the risk — and that is a defect you can detect mechanically by replaying the data, rather than a matter of editorial taste.
> The same criterion settles the perennial argument about documentation length. A summary needs whatever detail is required to stay honest at the precision claimed. Adding detail beyond that is optional; removing detail that makes it dishonest is not.

> **Worked example: the model card that read well and was wrong**
> A card reports overall AUC of 0.84 and states no fairness concern. Both statements are true of the portfolio in aggregate.
> Sliced by the declared operating boundary, however, the thin-file segment — 11% of volume, and inside the declared population — shows AUC 0.71 and an adverse impact ratio of 0.78. The card's claims are unsound: there exist in-boundary behaviours the summary excludes.
> The point is not that a human reviewer should have caught it. The point is that $b \sqsubseteq \gamma(s)$ *is a machine-checkable condition*, and a system that stores the evaluation data alongside the card can check every quantitative claim on every recompile.

4.3 Identity relative to a probe set

The Yoneda lemma expresses that an object is determined by its relationships to all others — an artefact is what it does. The totality of probes is unavailable, so we weaken it explicitly.

Definition 4.2 (Probe-relative equivalence). Let Π be a finite set of probes, each a pair of an input and a property assertion. Versions v_1, v_2 are Π-equivalent, $v_1 \equiv_\Pi v_2$, when their responses agree on every probe in Π within declared tolerance.

Proposition 4.3 (Version semantics). *Write $\Sigma(v) = (P, X, Y, \varphi)$ for the signature of a version. Then the three standard version increments have the following formal content:*

$$\begin{aligned} \textit{PATCH}: &\quad \Sigma(v') = \Sigma(v) \quad \textit{and} \quad v' \equiv_\Pi v, \\ \textit{MINOR}: &\quad \Sigma(v') = \Sigma(v) \quad \textit{with a different inhabitant of } P, \\ \textit{MAJOR}: &\quad \Sigma(v') \neq \Sigma(v). \end{aligned}$$

Moreover $\equiv_\Pi$ refines monotonically in Π: if $\Pi \subseteq \Pi'$ then $\equiv_{\Pi'} \subseteq \equiv_\Pi$, with equality of the artefacts in the limit of a separating probe set.

The monotonicity clause is the honest part, and it is the part that is systematically elided in practice. A claim of behavioural equivalence is exactly as strong as the probe set is rich; a governance system should therefore record Π with every such claim, measure its coverage, and treat an impoverished probe set as a deficiency rather than as a silent assumption.

In plain terms. You know two versions behave the same only to the extent that you actually tested them. That sounds obvious, and it is; what is not obvious is that most version-numbering schemes quietly assume otherwise.
A "patch" release is supposed to mean "nothing observable changed". Here that becomes a claim with a witness attached: nothing changed *with respect to this named set of tests.* If the test set is thin, the claim is thin, and the system should say so rather than let the version number imply more than was checked.
The Yoneda lemma is the limiting statement — test everything and the notion of "behaves the same" becomes genuine identity. We can never reach the limit, so we record how far along it we got.

Worked example: a patch release that was not one

A team refactors a scorecard's implementation from a hand-rolled scoring loop to a vectorised one and labels it 3.2.1 — a patch, no behavioural change. The regression suite is 400 rows sampled from recent applications, and it passes exactly.
What the suite does not contain is any application with a missing DSCR. The old implementation imputed the median before scoring; the new one propagates the null and the downstream binning maps it to the worst band. For roughly 2% of applications the score moves by 40 points, all in one direction.
The version number asserted equivalence. The probe set could not support the assertion, and nothing in the process required anyone to notice the gap between the two. Recording Π, measuring its coverage against the declared input domain, and treating thin coverage as a finding is the difference between a claim and a wish.

5 Indexing artefact kinds

Almost every structure in the domain is a family indexed by something else: versions by artefact, calibrations by version, deployments by environment, inventories by legal entity, and — critically — evidence requirements by artefact kind. The mathematics of indexed families is the fibration.

Let $p\colon \mathcal{E} \to \mathcal{B}$ be a fibration, $\mathcal{B}$ the category of artefact kinds, and the fibre $\mathcal{E}_c$ over kind c the structure required of artefacts of that kind: an evidence schema, a lifecycle, a default monitoring set, a document set, a contract template. Equivalently, by the Grothendieck construction, an indexed family $\mathcal{B}^{\mathrm{op}} \to \mathbf{Cat}$.

Proposition 5.1 (Conservative extension)**.** *Let $p\colon \mathcal{E} \to \mathcal{B}$ be a fibration and let $\mathcal{B}^+$ extend $\mathcal{B}$ by a new object c equipped with a chosen fibre $\mathcal{E}_c$, with no new morphisms into or out of existing objects. Let $p^+\colon \mathcal{E}^+ \to \mathcal{B}^+$ be the resulting fibration. Then the canonical inclusion $\mathcal{E} \hookrightarrow \mathcal{E}^+$ is full and faithful, commutes with the projections, and satisfies $\mathcal{E}^+_b = \mathcal{E}_b$ for every $b \in \mathcal{B}$. Consequently no judgement expressible over $\mathcal{B}$ is altered by the extension.*

Proof. By construction $\mathcal{E}^+$ is the coproduct $\mathcal{E} + \mathcal{E}_c$ over $\mathcal{B}^+ = \mathcal{B} + \{c\}$; fullness, faithfulness and fibrewise identity are immediate, and the cartesian liftings over $\mathcal{B}$ are unchanged since no morphisms were added. □

Proposition 5.1 is the resolution of pathology P2, and it is what licenses the claim that artefact kinds not yet invented are accommodated: supporting a new kind means supplying a fibre, and supplying a fibre disturbs nothing. The engineering counterpart is a plugin boundary rather than a schema column, with the totality of each fibre — that it supplies every required component — checkable at load time.

In plain terms. Think of a filing cabinet where each drawer holds one kind of artefact, and each drawer comes with its own set of forms: what must be filled in, who signs it, what gets checked, how often.
The bad design writes the list of drawers into the cabinet's frame, so adding a drawer means rebuilding the cabinet. The good design makes drawers independent, so adding one is just adding one. The proposition is the guarantee that adding a drawer cannot disturb the contents of any existing drawer — which is what lets you promise to support kinds of artefact that do not exist yet.

Worked example: adding a kind nobody anticipated

Suppose a bank begins using a quantum annealer for portfolio construction. It is not trained, not calibrated against market instruments, and not rule-based. Its parameters are an objective encoding and an annealing schedule; its outputs are stochastic and non-reproducible in the ordinary sense; the sensible evidence is distribution of solution quality against classical baselines, and stability across runs.
Under a schema that encodes kind as an enumeration, this requires a migration: a new enum value, new nullable columns for annealing parameters, changes to every validation screen that switches on kind, and a release. In practice this costs a quarter, so it does not happen, and the annealer is recorded as "Other" with its real characteristics in a free-text field — which is to say, it is outside governance while appearing to be inside it.
Under a fibration it is a package: an evidence schema, a lifecycle, a metric set, document templates, a contract template. Nothing existing is touched, and the metadata form the user sees is generated from the new schema. Proposition 5.1 is the guarantee that this is safe, not merely convenient.

6 Plural normative context

6.1 The problem

The same artefact may be outside the governed population of one regime, inside that of a second, classified high-risk by a third, and a financial-reporting key control under a fourth, simultaneously. These are not four values of one attribute. They are four logical systems, each with its own vocabulary, its own sentences, and its own notion of what makes a sentence true of an inventory state.

6.2 Institutions

An *institution* $\mathcal{I} = (\mathsf{Sign}, \mathsf{Sen}, \mathsf{Mod}, \models)$ consists of a category Sign of signatures, a functor $\mathsf{Sen}\colon \mathsf{Sign} \to \mathbf{Set}$ of sentences, a functor $\mathsf{Mod}\colon \mathsf{Sign}^{\mathrm{op}} \to \mathbf{Cat}$ of models, and a satisfaction relation $\models_\Sigma \subseteq |\mathsf{Mod}(\Sigma)| \times \mathsf{Sen}(\Sigma)$, subject to the *satisfaction condition*: for every signature morphism $\sigma\colon \Sigma \to \Sigma'$, every $M' \in |\mathsf{Mod}(\Sigma')|$ and every $\phi \in \mathsf{Sen}(\Sigma)$,

$$M' \models_{\Sigma'} \mathsf{Sen}(\sigma)(\phi) \quad\iff\quad \mathsf{Mod}(\sigma)(M') \models_\Sigma \phi.$$

Informally: truth is invariant under change of notation.

The interpretation is direct. A signature is the vocabulary a regime employs — complexity, exposure, purpose, application of statistical or economic theory, intended purpose, affected natural persons. A sentence is a normative statement in that vocabulary, of which the primary one is membership of the governed population. A model is the state of an inventory record as seen through that vocabulary. Satisfaction is compliance. A comorphism relates a regime institution to the core institution of the inventory itself.

Proposition 6.1 (Determination stability)**.** *Let $\varrho = (\Phi, \alpha, \beta)\colon \mathcal{I} \to \mathcal{I}'$ be an institution comorphism from a regime institution $\mathcal{I}$ to the core inventory institution $\mathcal{I}'$. Then for every* *core state* $M' \in |\mathsf{Mod}'(\Phi(\Sigma))|$ *and regime sentence* $\phi \in \mathsf{Sen}(\Sigma)$*,*

$$M' \models'_{\Phi(\Sigma)} \alpha_\Sigma(\phi) \quad \Longleftrightarrow \quad \beta_\Sigma(M') \models_\Sigma \phi.$$

Hence a determination computed on the core inventory using the translated obligation agrees with the determination computed natively in the regime's own vocabulary.

Proof. This is the defining condition of an institution comorphism. □

Corollary 6.2 (Falsifiable encoding test)**.** *Any observed disagreement between the two sides of Proposition 6.1 witnesses a defect in the encoding* (Φ, α, β) *rather than a genuine normative conflict. The condition is therefore a mechanically checkable property of a regime encoding.*

Corollary 6.2 is, we think, the practically significant observation of this section. Scope determinations are among the most consequential and least examined judgements in the domain: whether an artefact is inside a governed population determines whether any control applies to it at all. Encoding regimes as institutions converts the question "is this determination defensible?" from a matter of professional opinion into a derivation — sentence, evaluated facts, failing conjunct, citation — accompanied by a consistency property that can be tested against generated inventory states.

> **In plain terms.** Different regulators use different words for overlapping ideas, and mean different things by the same words. An institution is a way of writing down "a regulator's language", and a comorphism is a dictionary between that language and yours.
> The satisfaction condition is the demand that the dictionary be honest: translating a question and then answering it must give the same answer as answering it in the original language. That sounds like a technicality. It is actually the only mechanical check anyone has on whether a regulatory encoding is faithful, and it catches precisely the errors that produce a scope determination you cannot defend when asked.

> **Worked example: one artefact, four different answers, all correct**
> Consider a retrieval-augmented language model that drafts adverse-action explanation letters for declined credit applications.
> Under the 2026 US guidance it is *out of scope*: the definition covers complex quantitative methods applying statistical, economic or financial theory, and explicitly excludes generative systems. The determination is not "no" — it is a derivation whose failing conjunct is the generative-AI exclusion, with a footnote citation.
> Under the UK regime it is *in scope*: that definition admits qualitative outputs and expert-judgment inputs, and this artefact produces exactly that.
> Under the EU AI Act it is high-risk — not because of what it is, but because of what it is used for: it participates in a creditworthiness decision affecting a natural person.
> Under consumer-credit law it carries a further, quite specific obligation: the letter must state the actual principal reasons the application was declined.
> Four regimes, four different answers, none of them in conflict — because they are answers to four different questions asked in four different languages. Modelled as a boolean "in scope", this situation is unrepresentable and the system will force a wrong answer. Modelled as institutions, each answer is separately derived, separately citable, and separately testable.

Remark 6.3*.* The resolution of pathology P3 is then: admitting a new regime is exhibiting an institution and a comorphism. The core vocabulary, its schema, and every existing determination are untouched, by the same argument as Proposition 5.1. This is not a hypothetical benefit. During the preparation of this work the principal United States supervisory guidance in this domain, in force since 2011, was superseded by a revision that narrowed the definition of the governed population and excluded generative systems from it. Under a boolean or enumerated encoding of scope, that revision is a migration; under an institutional encoding it is a module.

7 Evidence over a semiring

7.1 Structure

Assurance evidence forms a directed acyclic derivation structure: a fitting evidence object supports a fitting run, which together with code and environment supports an artefact, which supports test outcomes, which support a validation conclusion, which supports an authorisation, which supports a deployment. Governance claims — *this version is validated for this use* — are derived from base evidence by conjunction (joint dependence) and disjunction (alternative derivations).

Annotate each base evidence item with an element of a commutative semiring $K = (K, \oplus, \otimes, 0, 1)$ and propagate with $\otimes$ for joint dependence and $\oplus$ for alternatives. This is exactly the setting of provenance semirings.

Proposition 7.1 (One computation, many questions)**.** *Let $\mathbb{N}[X]$ be the semiring of polynomials with natural coefficients over evidence identifiers X, and let a claim's derivation be evaluated in $\mathbb{N}[X]$. For any commutative semiring K and valuation $\nu\colon X \to K$ there is a unique semiring homomorphism $h_\nu\colon \mathbb{N}[X] \to K$ with $h_\nu \circ \iota = \nu$, and the evaluation of the claim in K equals h_ν applied to its $\mathbb{N}[X]$ evaluation.*

Proof. $\mathbb{N}[X]$ is the free commutative semiring on X; this is the universality result of Green, Karvounarakis and Tannen, restated. ☐

7.2 Seven questions, one engine

The consequence for the domain is that governance questions previously treated as distinct features are one computation under different valuations:

Semiring	$\oplus$, $\otimes$	**Governance question answered**
$\mathbb{B}$	$\vee$, $\wedge$	Is the claim supported at all?
$\mathbb{N}$	$+$, $\times$	How many independent derivations corroborate it?
Why(X)	$\cup$, pairwise $\cup$	Which minimal evidence sets suffice?
$\mathbb{N}[X]$	$+$, $\times$	Exactly how was it derived?
$([0,1], \max, \times)$	max, $\times$	With what confidence?
$(\mathbb{R}^+ \cup \{\infty\}, \min, +)$	min, $+$	At what least cost can a gap be closed?
security lattice	$\sqcap$, $\sqcup$	What classification does the conclusion inherit?
$(2^{\text{Regimes}}, \cap, \cup)$	$\cap$, $\cup$	For which regimes is the evidence admissible?
$(\text{Time}, \max, \max)$	max, max	As of when is the claim current?

The last two deserve comment. Regime admissibility is not decorative: evidence produced under one regime's methodology is not automatically admissible under another's, and a system that cannot express this will over-claim. Currency, evaluated in the max–max semiring, yields a definition of *staleness* that requires no heuristic: a derived document is stale exactly when the currency of its supporting evidence exceeds its own compile time.

> **In plain terms.** A semiring is just "a way of doing arithmetic" — something that plays the role of plus and something that plays the role of times.
> The insight is that tracking evidence has a fixed *shape*: this claim rests on that test AND that approval; this requirement can be met by this route OR that route. ANDs and ORs. If you keep the shape and swap the arithmetic, you get a completely different answer to a completely different question, from the same computation.
> Use true/false arithmetic and you learn whether the claim holds. Use probabilities and you learn how confident you should be. Use "smallest cost" arithmetic and you learn the cheapest way to fix a gap. Use sets-of-evidence and you learn what to put in front of an examiner. Nine features, one

implementation — and because they share a computation, they cannot contradict each other.

Worked example: one claim, five questions

Take the claim *version 3.2.1 is authorised for use in small-business origination.* Its derivation: three test results and a reproducibility check support a validation report, which together with either a committee approval or a delegated approval supports the authorisation.

- **Boolean:** `true`. The gate opens.
- **Why-provenance:** $\{\{\mathrm{VR}, \mathrm{APPR}_{\mathrm{cmte}}\}, \{\mathrm{VR}, \mathrm{APPR}_{\mathrm{deleg}}\}\}$ — two minimal routes. This is literally the answer to "what would you show an examiner?"
- **Trust** $([0,1], \max, \times)$**:** 0.91. One of the three tests was run on a stale sample and carries reduced trust, which propagates automatically.
- **Tropical** $(\min, +)$**:** 0 today. Had the validation report been missing, it would return 14.5 — the person-days on the cheapest path to closing the gap. That is a resourcing number generated from the evidence structure rather than estimated in a meeting.
- **Currency** $(\max, \max)$**:** `2026-08-11`. Compare against the model development document compiled on 2026-08-04 and the document is provably stale, with the specific superseding evidence identified.

Nobody wrote five features. Somebody wrote one traversal and five valuations, of about twenty lines each.

7.3 Consistency of independently produced evidence

Provenance records how a claim was derived but not whether independently produced evidence is mutually consistent. Treating evidence as a presheaf over the structure of an artefact — its components, slices and time windows — the question "do local assessments assemble into a global claim?" is the gluing condition of a sheaf, and the extent of failure is measurable as a consistency radius. Two validators assessing overlapping slices and reporting incompatible results on the overlap constitute a detectable defect that no per-slice check can see. This gives *effective challenge* — the domain's term for competent independent scrutiny — a global rather than merely additive character.

8 The order structure of risk

Classification of artefacts by risk is universally implemented as arithmetic on scores. It is more faithfully modelled as a monotone map between lattices.

Let M be a materiality lattice generated by a quantitative exposure chain joined with a qualitative purpose order, and C a complexity lattice, a product over its declared components (data quality, methodology, implementation integrity, intensity of use, interpretability, transparency, potential for bias). Let T be a finite chain of tiers.

Proposition 8.1 (Monotone classification)**.** *Let $\tau \colon M \times C \to T$ be monotone. Then increasing exposure, criticality of purpose, or complexity never lowers the assigned tier.*

Trivial as mathematics; consequential as an assurance property, since it is exactly the guarantee an examiner seeks and exactly what an unconstrained scoring formula fails to provide.

In plain terms. Two things drive how much scrutiny a model deserves: how much is riding on it, and how hard it is to get right. Most systems multiply these into a score and threshold the score, which quietly allows a very complex model with modest exposure to land in the same bucket as a simple model with enormous exposure — and, worse, allows the tier to *fall* when a number moves

the wrong way.
Keeping the two dimensions separate and requiring the classification to be monotone gives a guarantee you can state to a regulator in one sentence: *nothing we can learn about a model that makes it more consequential or more complicated will ever move it into a lighter control regime.* An unconstrained scoring formula cannot make that promise, and typically cannot even be checked.

Proposition 8.2 (Control adequacy)**.** *Let* Ctrl *be a lattice of control sets ordered by strictness, and let* $\mathrm{req}\colon T \to \mathrm{Ctrl}$ *and* $\sup\colon \mathrm{Ctrl} \to T$ *form a Galois connection,* $\mathrm{req}(t) \sqsubseteq c \iff t \sqsubseteq \sup(c)$*. Then the statements "the applied controls meet the tier's requirements" and "the tier is within what those controls can defend" are the same statement.*

The adjunction is doing real work. It means a control deficiency and an inflated classification are one defect viewed from two sides, and that a single definition answers both "what must be done for an artefact at this tier?" and "given what was actually done, what tier can this artefact legitimately claim?"

In plain terms. There are two questions people ask about controls, and they are usually answered by two different teams from two different documents.
"This is a Tier 1 model — what do we have to do?" and *"Here is what we actually did — is that good enough?"*
The adjunction says these are the same question. Write down one mapping and both answers follow, and they cannot drift apart. It also means a control gap and an over-optimistic tier are not two problems but one, seen from either end — which is why chasing them separately never converges.

9 Time

9.1 Bitemporality and causal admissibility

Fitting evidence assembled from historical records carries two independent time coordinates: the *valid time* at which a fact held in the world, and the *transaction time* at which the recording system learned it. Conflating them produces look-ahead leakage, which is the dominant silent failure mode in the domain: an artefact fitted on facts that had not yet been recorded when the decision was taken will assess well and perform badly, and the discrepancy will be attributed to degradation rather than to leakage.

Definition 9.1 (Causal admissibility)**.** Let $\mathcal{F}$ be a bitemporal store of facts, each carrying (t_v, t_τ). An assembly of fitting evidence for a decision at label time t_L, constructed at system time t_A, is *causally admissible* if it is a function only of facts in the down-set $\{\, \psi \in \mathcal{F} : t_v(\psi) \leq t_L \ \wedge \ t_\tau(\psi) \leq t_A \,\}$.

Proposition 9.2 (Characterisation)**.** *An assembly is causally admissible if and only if, for every feature* ϕ *and every row, the value used is*

$$\phi\big(e, t_L, t_A\big) \ = \ \textit{the value of } \phi \textit{ for } e \textit{ at } \max\{\, (t_v, t_\tau) : t_v \leq t_L \ \wedge \ t_\tau \leq t_A \,\}$$

under the lexicographic order on (t_v, t_τ)*.*

Proof. ($\Leftarrow$) Each value is by construction a function of a fact in the down-set, and the assembly is a function of these. ($\Rightarrow$) Suppose some value is not the maximal admissible one. Either it uses a fact outside the down-set, contradicting admissibility, or it uses a strictly earlier admissible fact, in which case the assembly is not a function of the down-set alone but of an arbitrary choice within it, and admissibility fails to determine it. $\square$

In plain terms. Every fact has two dates: when it was true, and when you found out. Ignore the second and you will build training sets containing information that did not exist at the moment the decision was made. The resulting model looks excellent in testing — it has been shown the answers — and performs badly in production.
This is the single most common serious defect in model development, it is invisible to every standard performance metric, and it is almost always misdiagnosed afterwards as "the model degraded".
The condition above is just: *use the most recent fact that was both true by the decision date and known by the assembly date.* Simple to state, easy to get wrong in SQL, and — crucially — checkable by a machine that recomputes it independently rather than trusting whoever wrote the query.

Worked example: the filed accounts that were restated

A borrower files 2026 Q1 accounts dated 31 March. They land in the warehouse on 20 May. The bank declines an application on 15 June. In August the borrower restates those accounts downward.
A training set assembled on 1 September, naively, joins on the accounts *as they now stand* — the restated figures. The model therefore learns from a version of the borrower's finances that did not exist on 15 June, and that in this case is worse, and therefore correlates suspiciously well with the decline. The model will show excellent discrimination in backtesting and will underperform in production, and the post-mortem will blame drift.
The correct assembly uses the figures with valid time on or before 15 June *and* transaction time on or before the assembly cut — the original filing, not the restatement.
Note the second thing bitemporality gives you. Because both clocks are recorded, the system can also answer the question that arises when the restatement lands: *which historical training sets, model versions and credit decisions used the superseded figures?* With one clock, that question is unanswerable.

Proposition 9.2 matters because it is *mechanically verifiable.* A governance system need not trust the author of an assembly query; it can independently recompute a stratified sample under the characterisation and assert agreement. The distinction between the two clocks additionally allows a restatement — "we discovered we were wrong" — to be distinguished from a genuine change in the world, and the affected assemblies, versions and historical decisions identified exactly.

9.2 Obligations

Normative obligations are temporal statements with deadlines and are naturally expressed in metric temporal logic, *e.g.*

$$\mathbf{G}\big(\,\text{tier} = 1 \;\rightarrow\; \mathbf{F}_{[0,\,365\text{d}]}\ \text{validated}\,\big), \qquad \mathbf{G}\big(\,\text{overlay active} \;\rightarrow\; \mathbf{F}_{[0,\,90\text{d}]}\ (\text{reviewed} \vee \text{expired})\,\big).$$

Runtime monitors for such formulae are synthesisable by standard constructions, so an obligation engine is a compiler rather than a scheduler with special cases. A thin deontic layer distinguishing obligation, permission and prohibition permits detection of normative contradiction — the simultaneous obligation and prohibition of the same proposition — before it reaches users, which arises more often than one would hope when several regimes apply at once.

In plain terms. Governance deadlines are statements of the form "whenever this becomes true, that must happen within so long". Written in this notation they can be compiled directly into monitors, so the obligation engine is a translator rather than a scheduler full of special cases. Adding an obligation is adding a line, not a code change.
The deontic layer keeps "must", "may" and "must not" distinct — which matters because when several regimes apply at once they occasionally demand contradictory things, and it is considerably

> better to discover that when the policy is written than when a user is blocked by it.

9.3 Documents as lenses

The relation between an evidence structure and a derived document is bidirectional: generated sections are computed from evidence, while narrative sections are authored and must survive regeneration. This is a well-behaved asymmetric lens with $\mathrm{get}\colon E \to D$ and $\mathrm{put}\colon E \times D \to E$ satisfying the standard laws. GetPut guarantees that regenerating an unedited document produces no spurious change; PutGet guarantees that an author's edit is not silently discarded. Staleness is then exactly the failure of $\mathrm{get}(e') = d$ for current evidence e', and the difference $\mathrm{get}(e') \ominus d$ identifies what changed.

> **In plain terms.** A document about a model has two kinds of content: numbers and tables generated from evidence, and prose written by a person. Regenerate carelessly and you destroy the prose; never regenerate and the numbers go stale.
> A lens is the standard tool for exactly this. Its laws are two promises: regenerating a document nobody edited changes nothing (so you get no spurious diffs and no churn), and an edit a person made is not silently thrown away on the next regeneration.
> And staleness stops being a judgement. A document is stale precisely when regenerating it would produce different generated content — and the difference tells the author exactly what to look at.

10 Laws as the criterion of seriousness

A foundation that is documented but unenforced decays into ornament within a small number of releases, and thereafter misleads. We therefore propose a discipline: each result above is restated as an executable law and checked by property-based testing against generated inputs, with failure treated as a build failure. Table 1 lists a representative set.

The adversarial entry deserves emphasis. A verifier for causal admissibility that silently ceases to detect leakage is a catastrophic and invisible regression; it must therefore be tested by injecting leakage that it is required to catch, not by examples it is known to pass.

11 Automation and its oracles

The constructions of Sections 4, 6 and 7 were introduced to secure governance properties. They have a second use, which was not the design intent and which we think is the most practically consequential consequence of the whole account: several of them are *decision procedures*, and a decision procedure is exactly what makes machine-generated output safe to accept.

This matters now because large language models and tool-using agents are being introduced into governance workflows at speed, and the prevailing control — "a competent human reviews the output" — is weak, expensive, and degrades precisely when the output is fluent. A formal substrate offers a better answer for a substantial fraction of the work.

11.1 Verified automation

Definition 11.1 (Oracle)**.** Let T be a task whose outputs lie in a set O. An *oracle* for T is a decidable predicate $\mathsf{ok}_T\colon O \to \mathbb{B}$, computable from the formal structure and from data independent of the output, such that $\mathsf{ok}_T(o)$ holds only if o is correct for T. A task is *oracle-backed* when such a predicate exists.

Definition 11.2 (Verified automation)**.** Given a task T with oracle ok_T and an arbitrary generator g — a heuristic procedure of any kind, including a stochastic one — *verified automation* of T is the composite that computes $g(x)$ and accepts it iff $\mathsf{ok}_T(g(x))$.

Law	Source	Test strategy
Determinism flag is correct	Section 2	Repeated sandboxed execution, differential comparison
Parameter accumulation	Proposition 2.5	Generated networks; assert composite parameter closure
Risk is lax, never strict	Corollary 3.5	Generated networks including the N_1/N_2 witness
Contract refinement on substitution	Section 4	Refinement checker at substitution time
Summary soundness	Proposition 4.1	Replay in-boundary data against every quantitative claim
Fibre totality	Proposition 5.1	Load-time validation of the kind registry
Satisfaction condition	Proposition 6.1	Generated inventory states $\times$ regime obligations
Provenance homomorphism	Proposition 7.1	Differential evaluation across all semirings
Classification monotonicity	Proposition 8.1	Generated lattice pairs
Control adequacy adjunction	Proposition 8.2	Exhaustive check over the finite chain
Causal admissibility	Proposition 9.2	Independent recomputation; adversarial leakage injection
Lens laws	Section 9	Round-trip tests per document template

Table 1: Results restated as executable laws.

Proposition 11.3 (Automation admissibility)**.** *If T is oracle-backed, the soundness of verified automation is independent of the generator: no incorrect output is accepted, whatever g is. The generator's error rate determines throughput, not correctness. If T is not oracle-backed, the correctness of the output is exactly the correctness of the generator.*

Proof. Immediate. In the first case acceptance implies ok_T, which implies correctness by Definition 11.1; the rejected fraction is $1 - \Pr[\mathsf{ok}_T(g(x))]$, a throughput quantity. In the second there is nothing between g and acceptance. $\square$

> **In plain terms.** If you can *check* an answer, it does not matter very much who or what produced it. A wrong answer is caught and discarded; the only cost is wasted effort. If you cannot check the answer, then trusting the answer is trusting the producer, and no amount of process changes that. So the useful question about automating a governance task is not "is the model good enough?" It is "do I have a check?" That question has a definite answer, and Table 2 gives it for the tasks this paper equips.

11.2 Which governance tasks are oracle-backed

The first row deserves emphasis because it inverts an apparent weakness. Section 6 makes regulatory extension cheap in principle, but encoding a forty-page supervisory statement into a signature and a set of sentences is expensive expert work, and that cost is the practical objection to the whole approach. It is also a task language models are unusually well suited to. And the output is checkable: a proposed encoding either satisfies Proposition 6.1 against generated inventory states or it does not. Generation is cheap, verification is mechanical, and the expert's role changes from authoring to adjudicating an encoding that has already passed a consistency test.

Task	Oracle	Source
Encode a regulatory regime	The satisfaction condition holds for the proposed comorphism	Proposition 6.1
Assert two versions behave alike	$\equiv_\Pi$ evaluated on the declared probe set	Proposition 4.3
Substitute one version for another	Contract refinement $\mathcal{K}' \preceq \mathcal{K}$	Section 4
Re-express an artefact in another representation	$\equiv_\Pi$ plus numerical tolerance	Proposition 4.3
Claim documentary support for an assertion	Boolean evaluation of the claim's derivation under the cited set	Proposition 11.4
Assemble fitting evidence without leakage	Causal admissibility	Proposition 9.2
Propose a remediation plan	*None needed* — the plan is computed in the tropical semiring	Section 7
Propose a probe	The probe executes and discriminates, or it does not	—
Propose a query	The query parses, type-checks and returns, or it does not	—

Table 2: Oracle-backed governance tasks and the construction that supplies the oracle.

11.3 Grounding verification is a semiring computation

The dominant failure mode of machine-generated governance text is the unsupported assertion. The standard mitigation — retrieval-augmented generation with citations — typically leaves the question "does this citation actually support this claim?" to a further model, which is to say unanswered. Over an annotated derivation structure it is a computation.

Proposition 11.4 (Citation soundness)**.** *Let c be a claim with derivation d_c over evidence identifiers X, and let $S \subseteq X$ be a cited set. Define $\nu_S\colon X \to \mathbb{B}$ by $\nu_S(x) = \top$ iff $x \in S$. Then S supports c if and only if the evaluation of d_c in $\mathbb{B}$ under ν_S is $\top$; equivalently, iff S contains some element of* $\mathrm{Why}(d_c)$.

Proof. By Proposition 7.1, evaluation in $\mathbb{B}$ under ν_S is h_{ν_S} applied to the $\mathbb{N}[X]$ evaluation of d_c. A polynomial evaluates to $\top$ under ν_S exactly when some monomial has all its variables in S, and the monomials of the $\mathbb{N}[X]$ evaluation are precisely the elements of $\mathrm{Why}(d_c)$. ☐

In plain terms. A generated sentence must come with the identifiers of the evidence it rests on. Checking it is then a matter of switching on exactly those identifiers and asking whether the claim still follows. If it does, the citation is genuine. If it does not, the sentence is rejected — not flagged for review, rejected.
The point is that no judgement is involved. Elsewhere, "is this citation relevant?" is itself a fallible model call. Here it is a Boolean evaluation over a structure that already exists for other reasons.

Worked example: a drafted paragraph, checked

A drafting assistant produces: *"Version 3.2.1 was approved for small-business origination following independent validation, which found discrimination within tolerance on all monitored slices."* It cites $S = \{\mathrm{VR}_{v2}, \mathrm{APPR}_{\mathrm{cmte}}\}$.
The claim *approved for this use* has derivation $(T_1 \cdot T_2 \cdot T_3 \cdot \mathrm{REP}) \cdot \mathrm{VR}_{v2} \cdot (\mathrm{APPR}_{\mathrm{cmte}} + \mathrm{APPR}_{\mathrm{deleg}})$. Under ν_S the test-result factors are $\perp$, so the product is $\perp$: the cited set does *not* support the claim as stated. The citation is incomplete, and the sentence is rejected with the missing identifiers named.
Note also the second clause — *"on all monitored slices"*. That is a quantified claim whose truth is not in the derivation at all; it is a statement about the slice set. It has no supporting monomial,

> and is therefore rejected as unsupported rather than published as plausible. This is the class of sentence that human review reliably misses, because it reads exactly like the rest.

11.4 The tasks that admit no oracle

Observation 11.5. Let $q = f(\Phi)$ be a derived governance quantity defined as the application of a versioned rule f to a fact set Φ — a risk classification, a control requirement, a scope determination. Then there is no automation question about computing q: evaluating f is deterministic and requires no generator. The automation question arises only for (i) supplying Φ, and (ii) proposing f.

Task (i) is oracle-backed whenever facts are sourced from systems of record, since the oracle is reconciliation against the source. Task (ii) is a normative choice: f *constitutes* the standard, so there is no independent specification against which a proposed f could be checked. It is exactly the kind of question that a person with authority is supposed to answer, and it is not made better by being answered faster.

> **In plain terms.** This reframes the question people usually ask. "Should we let a model assign risk tiers?" is malformed. Assigning a tier is not a generation task at all — it is arithmetic on a rule everyone has agreed to, and a spreadsheet could do it.
> The real questions are: where do the facts come from (checkable), and who chose the rule (a decision with accountability attached). Automating the second is not risky-but-tempting; it is a category error, because there is nothing for the automation to be checked against.

The same argument applies to concluding a validation, granting an approval, and accepting residual risk. These are not weakly-checkable tasks that we conservatively withhold from automation. They are tasks with no notion of correctness independent of the authority exercising them, and Proposition 11.3 therefore has nothing to offer.

11.5 Stratification: is a governing system that uses governed models circular?

If machine assistance is admitted into a system that governs machines, one should ask whether the account is well-founded.

It is, and the reason is stratification rather than good intentions. Let $\mathcal{G}$ denote the governing machinery — the classification map τ, the evidence structure, the institutions, the lifecycle categories. Let $\mathcal{A}$ denote the governed population. An assistant is registered as an element of $\mathcal{A}$: it has a parameter object (weights, prompt, corpus, tools), a fitting morphism (configuration), a contract, a classification, and evidence. None of $\mathcal{G}$ is an element of $\mathcal{A}$, and no construction in this paper requires a fixed point.

One dependency does cross the strata and is worth naming rather than hiding. If an assistant drafts a regime encoding (Table 2, first row), then part of $\mathcal{G}$ has been produced with help from an element of $\mathcal{A}$. The resolution is that the dependency is *mediated by an oracle* which is itself not machine-produced: the satisfaction condition is a property of institutions, proved in Proposition 6.1 and evaluated mechanically. Generation may cross the strata; acceptance may not.

> **In plain terms.** A system that governs models, using models it governs, sounds like it should collapse. It does not, provided one rule holds: *a machine may propose anything and decide nothing.* The proposing lives in the governed layer; the deciding lives in the governing layer; and where the two touch, a theorem sits between them.

12 A worked example

Consider a minimal estate of five artefacts. A is a discount curve constructed by solving against market instruments: parameter object the calibration set, fitting morphism a solver, deterministic given parameters. B is a swaption pricer consuming the curve and a volatility surface. C is a valuation-adjustment engine consuming the curve and the pricer's output. S is a statistically estimated scorecard on an unrelated portfolio. L is a foundation-model application drafting narrative from C's output and a document corpus.

Signatures. A is analytic in X but calibrated in P: P_A is inhabited daily by φ_A. S has P_S inhabited annually. L has $P_L = (\text{weights}, \text{prompt}, \text{corpus}, \text{tools})$ inhabited by configuration; its kernel is genuinely stochastic, so its determinism flag is false and its reproducibility evidence must pin the random source.

Composition and aggregate risk. The network copies A's output to both B and C. By Proposition 2.5 the composite has parameter object $P_C \otimes P_B \otimes P_A \otimes P_A$ — a shared factor. By Theorem 3.3 and Remark 3.4, the aggregate risk of $\{A, B, C\}$ strictly exceeds the join of their individual risks, and the interaction premium is attributable to the shared P_A. Recalibrating A is, by the corollary to Proposition 2.5, a change event for B, C and L. The scorecard S shares nothing and contributes no premium.

Plural scope. Under a narrow regime, A, B, C and S are in the governed population and L is excluded as generative. Under a broad regime all five are included, L by virtue of a definition admitting qualitative outputs. Under a product-safety regime, S alone is high-risk because it concerns natural persons. Proposition 6.1 guarantees that each determination is stable under the vocabulary in which it is evaluated, and Corollary 6.2 makes a defective encoding detectable. Crucially, L's exclusion from one population is recorded as a derivation with a failing conjunct and a citation, not as an absent row.

Evidence. The claim C *is authorised for its declared use* is derived from C's test outcomes, which depend on B's and A's current calibrations. Evaluated in $\mathrm{Why}(X)$ it returns the minimal evidence sets an examiner must be shown; in the max–max semiring it returns the currency of the oldest supporting item, which — since A recalibrates daily — immediately exposes any document whose figures predate the current calibration. Evaluated in the security lattice it propagates S's personal-data classification into every derived artefact, without bespoke code.

Observation. L is treated by contract: its assumption is the availability and provenance of the corpus, its guarantee is grounding and citation fidelity rather than accuracy in a statistical sense. A third-party artefact substituted for S would be opaque in the sense of Definition 2.2; only behavioural evidence would be obtainable, and its summary would be assessed for soundness by Proposition 4.1 rather than for completeness of development documentation, which is unobtainable.

Automation. Four tasks in this example are oracle-backed and may therefore be generated by any procedure whatever, including a stochastic one. The regime encodings for all three regulators are checkable by Proposition 6.1. A drafted paragraph describing C's authorisation is checkable by Proposition 11.4. A re-expression of S in a portable representation is checkable by $\equiv_\Pi$. A remediation plan for L's missing evaluation evidence is not generated at all — it is computed in the tropical semiring.

Two tasks are not oracle-backed and are not automation questions: what tier C should occupy, and whether L's residual risk is acceptable. By Observation 11.5 the first is arithmetic on an agreed rule and the second is an exercise of authority. Neither is improved by a generator.

The point of the example is that a single formal apparatus handles five artefacts that share no implementation technology, no fitting procedure, no output type, and no regulatory classification — and then tells us, without further argument, which of the work of governing them a machine may do.

13 Related work

Categorical foundations for learning. The **Para** construction and its use in gradient-based learning is due to Fong, Spivak and Tuyeras and elaborated by Cruttwell and collaborators; lenses supply the reverse pass. Our use of **Para** is orthogonal: we are not concerned with how parameters are optimised but with the fact that a parameter object exists and may be inhabited by procedures other than optimisation. Markov categories are due to Fritz and collaborators; representability, comparison of statistical experiments, and zero-one laws in that setting supply the technical background we draw on in Section 2.

Compositional systems and contracts. Assume–guarantee contract algebra is due to Benveniste and collaborators in the setting of cyber-physical system design; algebraic treatments of refinement, composition and quotient have been developed subsequently. We import the algebra unchanged and observe that operating boundaries and performance envelopes, terms used informally throughout the model governance literature, are exactly assumptions and guarantees.

Data provenance. Semiring provenance is due to Green, Karvounarakis and Tannen, with extensions to Datalog and description logics. Existing applications are to query answering and database annotation. Our application is to assurance evidence, where the derivations are governance inferences rather than relational operations and the semirings of interest include classification lattices and regime-admissibility sets.

Specification over multiple logics. Institution theory is due to Goguen and Burstall and has been applied to heterogeneous algebraic specification, hybrid logics and modelling-language semantics. We are not aware of prior application to overlapping normative regimes over a shared asset inventory, nor of the use of the satisfaction condition as a consistency test for regulatory encodings.

Machine learning metadata and documentation. Practical systems for model metadata, lineage tracking and registries address a subset of this territory but assume a homogeneous population of learned artefacts. Documentation proposals — model cards, datasheets, system cards — specify *what* a summary should contain; Proposition 4.1 contributes a criterion for *when* a summary is trustworthy that is independent of its contents.

Formal methods for machine learning. Abstract interpretation has been applied to neural network verification, yielding sound over-approximations of network behaviour. We use the same soundness notion at a different altitude: not to certify an artefact's behaviour but to certify that a governance summary does not understate it.

Language models, agents, and verified generation. The pattern of pairing an unreliable generator with a cheap checker is old — it is the generate-and-test structure of search, and the guess-and-verify structure of certifying algorithms — and has been revived for neural program synthesis, where a candidate program is accepted only if it passes a specification or test suite. Retrieval-augmented generation addresses grounding by supplying sources, but the question of whether a retrieved source *supports* a generated claim is generally answered by a further model call or by lexical overlap. Proposition 11.4 is a small observation with a useful consequence: over an annotated derivation structure that question is a Boolean evaluation, not an inference. Work on governance frameworks for generative and agentic systems in regulated settings is emerging rapidly; it is largely procedural, and Section 11 is intended as a complement — a criterion for *which* procedures need a human in them, derived from the structure rather than from caution.

Model risk regulation. The supervisory literature states the requirements this paper formalises but offers no formal apparatus, which is appropriate to its purpose. We take the prose seriously as a specification and observe that at least one of its central claims — concerning aggregate risk and common dependencies — has genuine mathematical content (Theorem 3.3) that has not, to our knowledge, been previously identified.

14 Limitations and falsification

No implementation study. The account is conceptual. We have not demonstrated that a system built on it is easier to construct, extend or operate than one built without it. That is the principal missing evidence, and we regard a longitudinal implementation study — particularly of the extension claims of Proposition 5.1 and Proposition 6.1 under real regulatory change — as the natural sequel.

Expressiveness. **Para**(**Stoch**) accommodates the population we have described, but stateful and interactive artefacts — simulation engines carrying internal state across invocations, agentic systems that take actions and observe consequences — are handled only by absorbing state into the input object, which is faithful but inelegant. A treatment in terms of coalgebra or open games may be more natural, and we do not claim to have settled the question.

Formalising normative text is lossy and contestable. Encoding a regime as an institution requires committing to a reading of regulatory prose that is deliberately open-textured. The encoding is a claim about the regulation, not the regulation, and different institutions may reasonably encode the same text differently. What the formalism supplies is that the claim is explicit, citable and testable for internal consistency — not that it is correct. We regard this as an improvement on prose in a spreadsheet, but it is not a resolution of legal interpretation.

Cost of abstraction. The apparatus imposes a comprehension cost on practitioners in a domain where mathematical sophistication is unevenly distributed. We have argued that each instrument earns this cost by delivering a property that would otherwise be hand-built or hand-checked; that argument is falsifiable, and a demonstration that any of the six delivers no property not obtainable more cheaply would properly remove it.

Oracles are necessary, not sufficient. Proposition 11.3 guarantees that no incorrect output is *accepted*; it says nothing about what an unchecked output does to the humans around it. There is a documented human-factors risk we can identify but not dissolve: machine-drafted governance text is fluent, fluency reads as quality, and reviewers of polished artefacts find fewer defects than reviewers of rough ones. An oracle that covers the citations of a document does not cover its rhetorical effect on the person attesting to it. We regard measuring reviewer edit distance — and treating a reviewer who changes nothing as a signal rather than a success — as a partial mitigation and not a solution.

The oracle boundary may be drawn too conveniently. Observation 11.5 classifies the decisions that matter most as admitting no oracle, which is a comfortable conclusion for anyone who would prefer humans to retain them. We believe the argument is sound — a rule that constitutes a standard cannot be checked against that standard — but note that it is exactly the sort of argument whose conclusion should be scrutinised in proportion to how much one likes it.

What would falsify the account. Three things. First, an artefact class in the domain that cannot be presented as a (P, φ) pair without distortion. Second, a fragility-sensitive aggregate risk measure that is nonetheless compositional — which Theorem 3.3 forbids, so a purported example would indicate an error in our failure semantics rather than in the theorem. Third, an oracle-backed task on which verified automation nonetheless produced worse outcomes than manual work, which would indicate that Proposition 11.3 abstracts away something load-bearing — most plausibly the tacit knowledge a person acquires *by* doing the task, which the composite discards. Fourth, and most likely, a demonstration that practitioners cannot use the derivations the formalism produces — that a scope determination presented as a failing conjunct with a citation is no more defensible in practice than a flag with a comment. The third is an empirical question about human factors and we cannot settle it from the armchair.

15 Conclusion

The management of heterogeneous model estates is treated as an engineering problem with a compliance flavour. We have argued that it is better understood as a problem with a missing foundation, and that the pathologies practitioners encounter — the training assumption, schema rigidity, scope as a boolean, evidence as assertion — are consequences of that absence rather than of insufficient effort.

The foundation we propose is modest in its ingredients and, we hope, not modest in its consequences. Defining a model as a parametric kernel makes artefacts that are never trained first-class rather than degenerate. Working in a Markov category rather than a Cartesian one makes shared dependency visible, and visible enough to prove that aggregate risk cannot be compositional — giving formal content to a supervisory expectation that has been stated in prose for fifteen years and implemented, so far as we can determine, nowhere. Indexing artefact kinds as fibres and normative regimes as institutions turns the two most common causes of re-architecture into module additions with conservative-extension guarantees. Annotating evidence over a semiring collapses seven governance questions into one computation. And restating each result as an executable law is what prevents the whole apparatus from becoming decoration.

There is a further consequence that we did not set out to find. Having built the structure for governance reasons, we discovered that it answers a question nobody asked it: *where may a machine do this work?* The criterion is not the model's capability, nor the task's sensitivity, but whether the structure supplies a decision procedure. Where it does — encoding a regulation, asserting behavioural equivalence, substituting a version, citing evidence for a claim — generation is free, because acceptance is checked. Where it does not, and in particular where the output would *constitute* the standard rather than answer to it, there is nothing for automation to be checked against, and the question is not one of risk appetite but of category.

We suspect this generalises. Wherever an organisation is deciding how much of a judgement-laden process to hand to a language model, the useful first question is not how good the model is. It is: what, in this domain, could ever tell us that the answer was wrong?

None of the mathematics is new. What we suggest is that a domain currently served by two families of tools that each solve half the problem is not waiting for a better product, but for a definition — and that the definition, once written down, turns out to pay for itself twice.

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